THE PUSHFORWARD ALONG A FLAT MORPHISM PRESERVES INJECTIVE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the adjunction and the transfer of the extension property are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Flat morphism of ringed topoi and exactness of the pullback. Same chapter, the definition of flatness of a morphism of ringed topoi and the exactness of the pullback it gives. Supplies the hypothesis and the exactness input.
- Injective module and the pullback-pushforward adjunction. Same chapter, the adjunction between f^* and f_* on modules, and the extension property of injective modules. Supplies the argument transferring injectivity.

Source reference. Original lemma 04JC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2267–2267. Navigation pointer invoked at proof line 2267; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015Z: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2269–2269. Navigation pointer invoked at proof line 2269; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. If f is flat, then $f_*\mathcal{I}$ is an injective $\mathcal{O}_\mathcal{D}$-module for any injective $\mathcal{O}_\mathcal{C}$-module $\mathcal{I}$.*

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Proof. In this case the functor f^* is exact, see Modules on Sites, Lemma 04JC. Hence the result follows from Homology, Lemma 015Z. □